

```

circle=[ (5,1) , (2,2) , (1,5) , (2,8) , (5,9) , (8,8) , (9,5) , (8,2) ]
def pat_circle():
    print('CIRCLE')
    for i in range(1,10):
        for j in range(1,10):
            if (i,j) in circle:
                print('+',end=' ')
            else:
                print(' ',end=' ')
        print()
pat_circle()

oval=[ (3,4) , (3,5) , (3,6) , (4,2) , (4,8) , (5,1) , (5,9) , (6,1) , (6,9) , (7,2) , (7,8) , (8,4) , (8,5) , (8,6) ]
def pat_oval():
    print('OVAL')
    for i in range(1,10):
        for j in range(1,10):
            if (i,j) in oval:
                print('+',end=' ')
            else:
                print(' ',end=' ')
        print()
pat_oval()

square=[ (2,2) , (2,3) , (2,4) , (2,5) , (2,6) , (2,7) , (2,8) , (3,2) , (3,8) , (4,2) , (4,8) ,
          (5,2) , (5,8) , (6,2) , (6,8) , (7,2) , (7,8) , (8,2) , (8,3) , (8,4) , (8,5) , (8,6) , (8,7) , (8,8) ]
def pat_square():
    print('SQUARE')
    for i in range(1,10):
        for j in range(1,10):
            if (i,j) in square:
                print('+',end=' ')
            else:
                print(' ',end=' ')
        print()
pat_square()

rhombus=[ (1,3) , (1,4) , (1,5) , (1,6) , (1,7) , (3,2) , (3,6) , (5,1) , (5,2) , (5,3) , (5,4) , (5,5) ]
def pat_rhombus():
    print('RHOMBUS')
    for i in range(1,10):
        for j in range(1,10):
            if (i,j) in rhombus:
                print('+',end=' ')
            else:
                print(' ',end=' ')
        print()
pat_rhombus()

triangle=[ (2,5) , (3,4) , (3,6) , (4,3) , (4,7) , (5,2) , (5,3) , (5,4) , (5,5) , (5,6) , (5,7) , (5,8) ]
def pat_triangle():
    print('TRIANGLE')
    for i in range(1,10):
        for j in range(1,10):
            if (i,j) in triangle:
                print('+',end=' ')
            else:
                print(' ',end=' ')
        print()
pat_triangle()

```

```

Semi_circle=[(4,4),(4,5),(4,6),(4,7),(5,3),(5,8),(6,2),(6,9),(7,2),(7,9),
             (8,2),(8,3),(8,4),(8,5),(8,6),(8,7),(8,8),(8,9)]
def pat_Semi_circle():
    print('SEMI_CIRCLE')
    for i in range(1,10):
        for j in range(1,10):
            if (i,j) in Semi_circle:
                print('+',end=' ')
            else:
                print(' ',end=' ')
        print()
pat_Semi_circle()

Rectangle=[(1,2),(1,3),(1,4),(1,5),(1,6),(1,7),(1,8),(1,9),(2,2),(2,9),
            (3,2),(3,9),(4,2),(4,3),(4,4),(4,5),(4,6),(4,7),(4,8),(4,9)]
def pat_Rectangle():
    print('RECTANGLE')
    for i in range(1,10):
        for j in range(1,10):
            if (i,j) in Rectangle:
                print('+',end=' ')
            else:
                print(' ',end=' ')
        print()
pat_Rectangle()

trapezoid=[(2,3),(2,4),(2,5),(2,6),(2,7),(3,2),(3,8),(4,1),(4,9),(5,1),
            (5,2),(5,3),(5,4),(5,5),(5,6),(5,7),(5,8),(5,9)]
def pat_trapezoid():
    print('TRAPEZOID')
    for i in range(1,10):
        for j in range(1,10):
            if (i,j) in trapezoid:
                print('+',end=' ')
            else:
                print(' ',end=' ')
        print()
pat_trapezoid()

pentagon=[(1,5),(3,3),(3,7),(4,2),(4,8),(6,3),(6,7),(7,4),(7,5),(7,6)]
def pat_pentagon():
    print('PENTAGON')
    for i in range(1,10):
        for j in range(1,10):
            if (i,j) in pentagon:
                print('+',end=' ')
            else:
                print(' ',end=' ')
        print()
pat_pentagon()

```

```

hexagon=[ (2,4) , (2,5) , (2,6) , (3,3) , (3,7) , (4,2) , (4,8) , (5,1) , (5,9) , (6,2) , (6,8) , (7,3) , (7,7) ,
(8,4) , (8,5) , (8,6) ]
def pat_hexagon():
    print('HEXAGON')
    for i in range(1,10):
        for j in range(1,10):
            if (i,j) in hexagon:
                print('+',end=' ')
            else:
                print(' ',end=' ')
        print()
pat_hexagon()
heptagon=[ (1,5) , (2,3) , (2,7) , (3,2) , (3,8) , (4,2) , (4,8) , (5,2) , (5,8) , (6,3) , (6,7) , (7,4) , (7,5) , (7,6) ]
def pat_heptagon():
    print('HEPTAGON')
    for i in range(1,10):
        for j in range(1,10):
            if (i,j) in heptagon:
                print('+',end=' ')
            else:
                print(' ',end=' ')
        print()
pat_heptagon()
octagon=[ (2,4) , (2,5) , (2,6) , (3,3) , (3,7) , (4,2) , (4,8) , (5,2) , (5,8) , (6,2) , (6,8) , (7,3) , (7,7) , (8,4) , (8,5) , (8,6) ]
def pat_octagon():
    print('OCTAGON')
    for i in range(1,10):
        for j in range(1,10):
            if (i,j) in octagon:
                print('+',end=' ')
            else:
                print(' ',end=' ')
        print()
pat_octagon()
nonagon=[ (1,5) , (2,3) , (2,7) , (4,1) , (4,9) , (6,1) , (6,9) , (8,3) , (8,7) , (9,4) , (9,5) , (9,6) ]
def pat_nonagon():
    print('NONAGON')
    for i in range(1,10):
        for j in range(1,10):
            if (i,j) in nonagon:
                print('+',end=' ')
            else:
                print(' ',end=' ')
        print()
pat_nonagon()
decagon=[ (1,4) , (1,5) , (1,6) , (2,3) , (2,7) , (3,2) , (3,8) , (4,2) , (4,8) ,
(5,2) , (5,8) , (6,2) , (6,8) , (7,3) , (7,7) , (8,4) , (8,5) , (8,6) ]
def pat_decagon():
    print('DECAGON')
    for i in range(1,10):
        for j in range(1,10):
            if (i,j) in decagon:
                print('+',end=' ')
            else:
                print(' ',end=' ')
        print()
pat_decagon()

```

```

heart=[(2,2),(2,3),(2,7),(2,8),(3,1),(3,4),(3,6),(3,9),(4,1),
        (4,9),(5,2),(5,8),(6,3),(6,7),(7,4),(7,6),(8,5)]
def pat_heart():
    print('HEART')
    for i in range(1,10):
        for j in range(1,10):
            if (i,j) in heart:
                print('+',end=' ')
            else:
                print(' ',end=' ')
        print()
pat_heart()

```

CIRCLE

```

          +
        +       +
      +           +
    +               +
  +                   +
+                       +
+                       +
+                   +
+               +
+           +
+       +
+   +
+

```

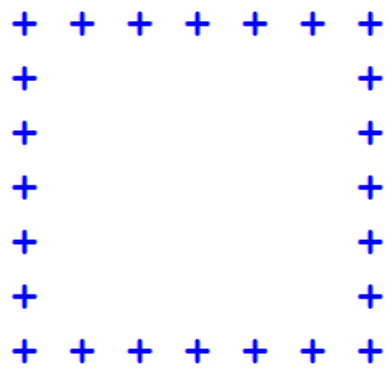
OVAl

```

          + + +
        +   +   +
      +       +
    +           +
  +               +
+                   +
+                   +
+               +
+           +
+       +
+   +
+

```

SQUARE



RHOMBUS



TRIANGLE



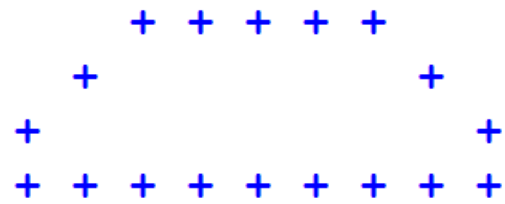
SEMI_CIRCLE



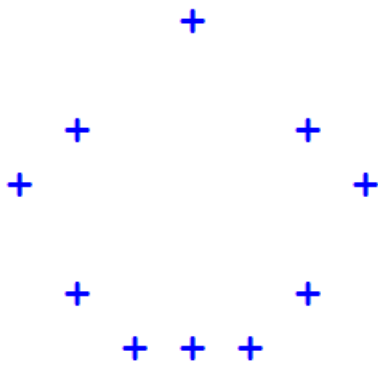
RECTANGLE



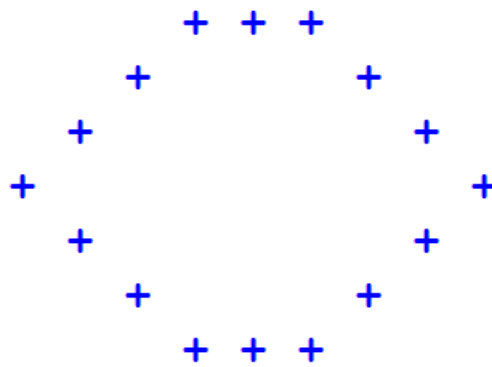
TRAPEZOID



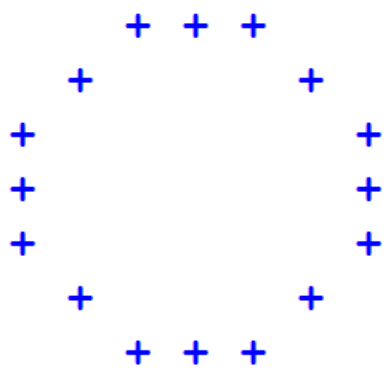
PENTAGON



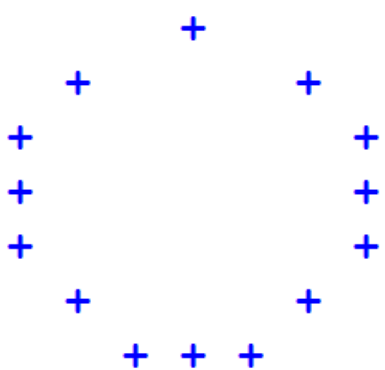
HEXAGON



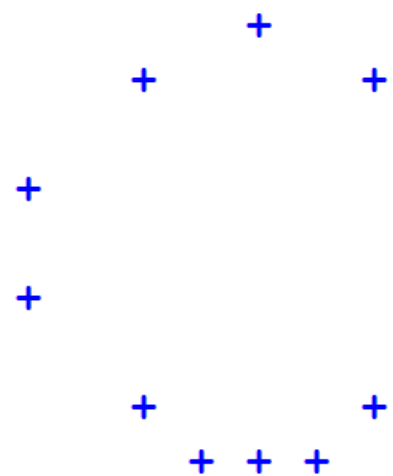
OCTAGON



HEPTAGON



NONAGON



DECAGON



HEART

